

Reading the Dataset

Credit Card Approval Prediction System | Epic 3: Dataset Download and Understanding

Objective

Load the two raw datasets — application records and credit history — into pandas DataFrames for exploration and target variable derivation.

Two Dataset Files

File	Records	Columns	Description
Dataset/application_record.csv	438,557	18	Applicant demographics, financial profile, and contact flags
Dataset/credit_record.csv	1,048,575	3	Monthly credit STATUS per applicant. One row per applicant

Code: Loading Both Datasets

```
import pandas as pd

application_df = pd.read_csv('Dataset/application_record.csv')
credit_df      = pd.read_csv('Dataset/credit_record.csv')

print('Application shape:', application_df.shape) # (438557, 18)
print('Credit shape:',      credit_df.shape)      # (1048575, 3)

application_df.head()
credit_df.head()
```

application_record.csv — Columns

Column	Type	Description
ID	int	Unique applicant identifier (join key)
CODE_GENDER	str	M / F
FLAG_OWN_CAR	str	Y / N — owns a car
FLAG_OWN_REALTY	str	Y / N — owns real estate
CNT_CHILDREN	int	Number of children
AMT_INCOME_TOTAL	float	Annual income
NAME_INCOME_TYPE	str	Working / Commercial Associate / Pensioner / State Servant / Student
NAME_EDUCATION_TYPE	str	Academic Degree / Higher Education / Incomplete Higher / etc.
NAME_FAMILY_STATUS	str	Married / Single / Civil Marriage / Separated / Widow
NAME_HOUSING_TYPE	str	House/Apartment / Rented / With Parents / etc.

Column	Type	Description
DAYS_BIRTH	int	Negative days relative to application date (age)
DAYS_EMPLOYED	int	Negative days employed; 365243 = unemployed/retired
FLAG_MOBIL / WORK_PHONE / PHONE / EMAIL	ID/1	contact flags
OCCUPATION_TYPE	str	134,203 missing values — filled with 'Unknown'
CNT_FAM_MEMBERS	float	Total family members including applicant

credit_record.csv — Columns

Column	Description	Values
ID	Applicant ID (join key)	Matches application_record ID
MONTHS_BALANCE	Month offset (0=current, -1=last month)	Negative integers
STATUS	Payment status for that month	0=1-29 days late, 1=30-59 days, 2=60-89 days, 3=90-119 days, 4=120-149 days